Foundations of Generative AI (ICT4403) — Assignment 2
Autoregressive Next-Word Text Generation

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1 Problem statement and objective

Develop and compare four autoregressive (next-word) text-generation models — Simple RNN, Bidirectional RNN, LSTM, and GRU — on a chosen textual corpus. Each model predicts w_{t+20} from context $[w_t, \ldots, w_{t+19}]$. Performance is evaluated using loss, perplexity, prediction accuracy, and training time, as well as qualitative coherence, diversity, contextual consistency, and repetition of generated text, under three sampling temperatures $T \in \{0.5, 1.0, 1.5\}$.

2 Dataset description and source

- **Dataset:** *The War of the Worlds*, H. G. Wells (public domain).
- **Source:** `https://www.gutenberg.org/files/36/36-0.txt` (Project Gutenberg eBook #36).
- Tokens after cleaning: ∼70k; vocabulary ∼3.1k (minimum frequency 2).
- Split: 80/10/10 train/validation/test; context length 20 (many-to-one target avoids future-token leakage for the Bidirectional model).

3 Models and experimental conditions

All models share: embedding dim 128, hidden dim 256, 1 layer, dropout 0.2, Adam (lr = 10^{-3}), gradient clipping 1.0, CrossEntropyLoss, seed 42. The Bidirectional model splits the hidden size across directions.

$$\text{PPL} = \exp(\mathcal{L}), \qquad P(w) \propto \exp(z_w/T).$$

4 Quantitative results

Model	Params	Val PPL	Test PPL	Test Acc	Time (s)
SimpleRNN	1293086	108.55	96.86	0.2036	91.1
BiRNN	1260318	110.90	96.15	0.2155	121.1
LSTM	1589534	99.75	**86.72**	**0.2252**	211.9
GRU	1490718	100.08	87.13	0.2209	165.1

5 Model comparison

- **LSTM** achieves the lowest validation/test perplexity and accuracy.
- **GRU** is close (87.13 vs 86.72 PPL) while training ∼22% faster and uses fewer parameters.
- **BiRNN** showed only marginal benefit because predictions are causal (left context).
- **Simple RNN** trains fastest but underfits longer dependencies.

6 Sampling-temperature analysis

Text was sampled from identical seeds at $T = 0.5, 1.0, 1.5$ (outputs in `results/generated_samples.txt`).

- $T = 0.5$: conservative and repetitive (high <unk> repetition), most grammatically stable.
- $T = 1.0$: best balance of coherence and diversity; recommended default.
- $T = 1.5$: highest diversity but frequent drift, odd word choices and punctuation.

7 Conclusion

LSTM is the best-performing model on held-out perplexity and accuracy. **GRU** is effectively tied and more efficient, so it is the recommended practical choice for iterative autoregressive generation on this corpus. Simple RNN and Bidirectional RNN remain viable but less accurate.

8 Submission details

- Student: Madhavan Komandur; Registration 230953298.
- Dataset: The War of the Worlds (H. G. Wells) — `https://www.gutenberg.org/files/36/36-0.txt`.
- GitHub: `https://github.com/sltrtn/FGAI-Assignment2-Autoregressive-TextGeneration`.
- Code, plots, metrics and generated samples: `text_generation.ipynb` and `experiment.py`.